MARIUM YOUSUF

Curriculum Vitae

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RESEARCH VISION

Develop robust data-driven frameworks for understanding the complex dynamics of the brain, with a particular emphasis on modeling functional and effective connectivity in neural systems.

EDUCATION

Ph.D. in Applied Mathematics (Ph.D. Candidate) — expected Summer 2026

M.S. in Applied Mathematics — Aug 2023

M.S. in Computer Science — May 2022

University of Arizona, Tucson AZ

B.S. in Mathematical Sciences, *summa cum laude* — Dec 2017

Northern Illinois University, DeKalb IL

Associates in Science — May 2015

McHenry County College, Crystal Lake IL

ACADEMIC APPOINTMENTS

Ph.D. Student, University of Arizona (UArizona)

Research

Department of Mathematics — Fall 2022-Present

Developing a probabilistic graphical model to extract neural effective connectivity from spike trains.
Keywords: Replay, Brain Connectivity, Causal Discovery, Network Inference, Stochastic Modeling
Advisors: Jean-Marc Fellous, Michael Chertkov

Department of Computer Science — Fall 2019-Spring 2022

Implemented an approximation of G-Wishart marginal likelihood to learn sparse graphical structures representing different levels of functional brain connectivity.

Primary Instructor

Elements of Calculus, Dept. of Mathematics — Summer 2024, Summer 2025
College Algebra, Dept. of Mathematics — Spring 2024, Spring 2025
Calculus Preparation, Dept. of Mathematics — Fall 2023

Teaching Assistant

Understanding Data, Dept. of Mathematics — Spring 2023
College Algebra, Dept. of Mathematics — Fall 2022
Discrete Data Structures, Dept. of Computer Science — Fall 2019, Yr. of 2021

Graduate Research Aide, Argonne National Laboratory — Summers 2021-2023

Automated high-throughput TEM and X-ray mouse brain image processing by integrating visualization systems (WebKnossos, NeuroGlancer) and reducing reliance on manual tools like TrakEM2.

Graduate Research Assistant, Missouri University of Science and Technology

Department of Computer Science — Fall 2018-Summer 2019

Pre-processed and analyzed data collected from dementia patients to infer the role of sedentary body movements in early diagnosis of dementia.

Pre-Doctoral Intern, Argonne National Laboratory — Apr-Aug 2018

Research Aide Built Python tools for efficient visualization of real-time data from sensors located in Chicago for an Array of Things project.

Lecturer Prepared Python and Jupyter Notebook materials and lectured in a three-day Big Data Visualization Camp for rising high-school seniors.

Undergraduate Research Aide, Argonne National Laboratory — Summer 2017

Configured Apache Spark in Jupyter Notebook to analyze real-time simulated data for visualization tasks.

Undergraduate Teaching Assistant, Northern Illinois University (NIU)

UNIX and Networking, Dept. of Computer Science — Fall 2017

MANUSCRIPTS

Yousuf, M., Pagnier, L., Fellous, J.M., Chertkov, M. 2026. *Inferring Directed Functional Connectivity Graphs from Neural Spike Trains.* Manuscript under review at Neural Computation. — March 2026

CONFERENCE PRESENTATIONS

Yousuf, M., Pagnier, L., Chertkov, M., & Fellous, J.-M. (2026). *Causality in replay: Detecting effective connectivity from spike trains.* Poster presented at Dynamics Days, Tucson, AZ, Jan 9–11.

Yousuf, M., Pagnier, L., Chertkov, M., & Fellous, J.-M. (2025). *Causality in replay: Detecting effective connectivity from spike trains.* Poster presented at Neuroscience 2025 (Society for Neuroscience), San Diego, CA, Nov 15–19.

Yousuf, M., Pagnier, L., Chertkov, M., & Fellous, J.-M. (2025). *Causality in replay: Comparing methods to detect effective connectivity from spike trains.* Poster presented at NITMB MathBio Convergence Conference, Chicago, IL, Aug 11–18.

Yousuf, M., Chertkov, M., & Fellous, J.-M. (2024). *Hippocampal replay and sleep's hidden language: Methods for detecting functional connectivity from spike trains.* Poster presented at Neuroscience 2024 (Society for Neuroscience), Chicago, IL, Oct 5–9.

INVITED AND LOCAL PRESENTATIONS

Yousuf, M. (2026). *Investigating the success of multiple-choice assessments in a math classroom.* Invited presentations at CIRTL Network Teaching-as-Research Presentations (virtual), Apr 8; and at College Teaching ShareFest (virtual), Apr 17.

Yousuf, M. (2025). *Investigating the success of multiple-choice assessments in a math classroom.* Poster presented at the Graduate Interdisciplinary Programs Student Research Showcase, University of Arizona, Tucson, AZ, Dec 11.

Yousuf, M. (2025). *Causality in Hippocampal Replay: Detecting Functional Connectivity from Spike Trains.* Invited presentation at the Human Augmented Analytics Group Group, Georgia Institute of Technology, Atlanta, GA.

Yousuf, M. (2024). *Hippocampal replay and sleep's hidden language: Methods for detecting functional connectivity from spike trains.* Contributed talk at the Arizona Women's Symposium in Mathematics, Flagstaff, AZ, Sep 21.

Yousuf, M. (2024). *Detecting replay in multi-unit spiking data: Bayesian networks.* Invited poster presentation at the Graduate Interdisciplinary Programs Student Research Showcase, University of Arizona, Tucson, AZ, Dec 6.

Yousuf, M., Chertkov, M., & Fellous, J.-M. (2023). *Detecting replay in multi-unit spiking data: Bayesian networks.* Poster presented at Neuroscience 2023 (Society for Neuroscience), Washington, DC, Nov 11–15; and at the Arizona Women's Symposium in Mathematics, Flagstaff, AZ, Nov 19.

Yousuf, M. (2019). *Diagnosing Dementia Using Smart Chair, Wearable Device, and Testing Apps.* Presentation at the Annual Graduate Research Symposium, Intelligent Systems Center, Rolla, MO, Apr 3.

AWARDS, HONORS, AND SCHOLARSHIPS

Trainee Professional Development Award (1000 USD), Society for Neuroscience	Nov 2025
Grogan Scholarship Award (6000 USD), Dept. of Mathematics, UArizona	Fall 2025
Herbert E. Carter Travel Award (600 USD, 100 USD), Graduate College UArizona	2024, 2023
TA of the Month, Dept. of Computer Science, UArizona	Oct 2021
Grace Hopper Student Scholar	Oct 2019
Norma K. Stelford Mathematics Endowment, NIU (graduating senior in mathematics with the highest GPA)	Dec 2017
The Clarence Ethel Hardgrove Mathematics Scholarship, NIU (incoming transfer with excellent prior record in mathematics)	2015-2016
International Undergraduate Scholarship, NIU	2015-2017

ADDITIONAL ACADEMIC TRAINING

UArizona NSF Research Training Group — Fall 2024, Fall 2025

Funded through NSF-supported research group focused on modern computational methods for data-driven modeling and applications.

Simons Laufer Mathematical Sciences Institute Summer 2025

Selected to attend a summer graduate workshop on Local Limits of Random Graphs held at Université Paris-Saclay Mathematics Institute in France.

Center for the Integration of Research, Teaching, and Learning (CIRTL) Fall 2021-Fall 2025

Completed all designations (*Associate, Practitioner, & Scholar*) in the CIRTL's three-tiered teaching certificate program, with training in evidence-based undergraduate STEM teaching.

PROFESSIONAL DEVELOPMENT

Mentor, STAR Lab, UArizona 2024-2026

Mentored high school seniors in conducting data-driven scientific research

Chapter Officer, SIAM UArizona Chapter 2024-2026

Served as a Treasurer and Vice President

Secured 490 USD funds from the SIAM board and assisted in planning chapter events

Co-organized SIAM mini-conference for graduate students from diverse disciplines

Panelist, Graduate Teaching Assistants' Orientation and Training, UArizona Aug 2024

Invited to serve in GTA training for incoming graduate students in the Dept. of Mathematics

Mentor, Undergraduate Mathematical Modeling, UArizona Spring 2024

Mentored an undergraduate team for a capstone project on learning languages using Markov Chains

Volunteer, Outreach Program BASIS Oro Valley High School, Oro Valley AZ Mar 2023, 2024

Brain- and memory-inspired educational activities for 6th-graders

Mentee, Women in STEM Mentorship Program, University of Arizona 2020-2021

Developer, Healthcare Hackathon 2019. Codefi, Cape Girardeu MO July 2019

Developer, AI Chess Tournament. Sandia National Laboratory, Rolla MO May 2019

Developer, PickHacks Hackathon. Major League Hacking, Rolla, MO. 2018

Volunteer, Hopper for Grace Hopper Conference 2018